



NANYANG TECHNOLOGICAL UNIVERSITY

EE6407 Genetic Algorithms & Machine Learning

Assignment 1 & 2 Report

MSc in Electrical and Electronic Engineering

Yan Ziming

G2507084J

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1 Introduction

For the course EE6407, I implement two assignments, including the Naïve Bayes Classifier and the Fisher Linear Discriminate Classifier. This report summarizes the methods and results of these two assignments. I use Python as the programming language and Jupyter Notebook as the development environment. The followings show the details of my design and implementation.

2 Assignment 1

2.1 Handling Missing Values and Outliers

2.1.1 Missing values

Detection:

For the missing value detection, I use the pandas library to check, with `isna()` function. This function would return a matrix reflects if the content in each position is empty or not. By summing the values in each column, I can determine the number of times each feature's missing values occur 1.

```
missing_train=Train_data.isna()  
missing_train_n=missing_train.sum()
```

Figure 1: Missing Value Detection Code

At first, there seems no missing values. however, after trying to transform datatype from dtype to float, there seems several data cannot successfully transformed, leading to new missing values. The missing value detection for training set consequence:

Feature	Missing Value Count
A	2
B	0
C	0
D	1
E	0

Handling:

After detecting the missing values, there are

2.1.2 Outliers

Detection:

abcd

Handling:

2.2 Design a Naïve Bayes classifier

```
class GNB_Classifier:
    def __init__(self):
        pass

    def fit(self,X,y):
        self.classes=np.unique(y)
        self.means=[]
        self.vars=[]
        self.priors=[]
        for w in self.classes:
            X_w=X[y==w]
            prior_w=X_w.shape[0]/X.shape[0]
            mean_w=np.mean(X_w,axis=0)

# cov_w=np.cov(X_w,rowvar=False)+1e-6*np.eye
(X.shape[1])
            var_w = np.var(X_w, axis=0)
            self.priors.append(prior_w)
            self.means.append(mean_w)
            # self.vars.append(cov_w)
            self.vars.append(var_w)

        self.priors = np.array(self.priors)
        self.means = np.array(self.means)
        self.vars = np.array(self.vars)

    def ConditionPwithGaussian(self,x,mean,
var):
        eps = 1e-6
        coeff = 1.0 / np.sqrt(2 * np.pi *
var + eps)
        exponent = - (x - mean) ** 2 / 2 *
var + eps)
        return coeff * np.exp(exponent)
```

```

def predict(self, X):
    X = np.array(X)
    y_pred = []

    for x in X:
        posteriors = []
        for i, c in enumerate(self.classes):
            prior_log = np.log(self.priors[i])

            likelihood_log = np.sum(np.log(
self.ConditionPwithGaussian(x, self.means[i]
], self.vars[i])))
            posteriors.append(prior_log +
likelihood_log)
            y_pred.append(self.classes[np.argmax
(posterior)])

        return np.array(y_pred)

def evaluate(self, X, y):
    y = np.array(y)
    y_pred = self.predict(X)
    accuracy = np.mean(y_pred == y)
    return accuracy

```

Figure 2: Code of Classifier Establishment

2.3 Prediction and Analysis

Table 1: Results of Gaussian Naive Bayes Classifier

Metric	Value
Training Accuracy	0.958
Predicted Labels	[1 1 1 1 1 1 1 1 1 3 2 2 2 2 2 2 2 2 2 3 3 3 3 3 3 3]

3 Assignment 2

3.1 Fisher Linear Discriminate Classifier

Establish:

Training:

3.2 Prediction and Analysis

Table 2: Results of Fisher Linear Discriminant Analysis (LDA)

Metric	Value
Predicted Labels	-1 -1 -1 -1 -1 -1 -1 -1 -1 1 1 1 1 1 1 1 1 1 1 1 1
w	0.001786 0.003473 0.001947 0.001683 0.002658
b	-0.008700
Training Accuracy	0.932

4 Conclusion